Source “Vit” Vittinghoff, E. (2012). Regression Methods in Biostatistics: Linear, Logistic, Survival, and Repeated Measures Models (2nd ed. 2012). Springer Nature. https://doi.org/10.1007/978-1-4614-1353-0

Table 4.2 Adjusted regression of glucose on exercise

```
. regress glucose exercise age drinkany BMI if diabetes == 0

      Source |       SS       df       MS              Number of obs =    2028
-------------+------------------------------           F(  4,  2023) =   39.22
       Model |  13828.8486     4  3457.21214           Prob > F      =  0.0000
    Residual |  178319.973  2023  88.1463042           R-squared     =  0.0720
-------------+------------------------------           Adj R-squared =  0.0701
       Total |  192148.822  2027  94.7946828           Root MSE      =  9.3886

------------------------------------------------------------------------------
     glucose |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
    exercise |   -.950441     .42873    -2.22   0.027    -1.791239   -.1096426
         age |   .0635495   .0313911     2.02   0.043     .0019872    .1251118
    drinkany |   .6802641   .4219569     1.61   0.107    -.1472513     1.50778
         BMI |    .489242   .0415528    11.77   0.000     .4077512    .5707328
       _cons |   78.96239   2.592844    30.45   0.000     73.87747    84.04732
------------------------------------------------------------------------------
```


Table 4.4 Regression of physical activity on glucose

```
. regress glucose i.physact if diabetes == 0

      Source |       SS       df       MS              Number of obs =    2032
-------------+------------------------------           F(  4,  2027) =    4.43
       Model |  1673.09022     4  418.272554           Prob > F      =  0.0014
    Residual |  191344.609  2027  94.3979322           R-squared     =  0.0087
-------------+------------------------------           Adj R-squared =  0.0067
       Total |  193017.699  2031  95.0357946           Root MSE      =  9.7159

------------------------------------------------------------------------------
     glucose |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
     physact |
           2 |  -.8584489   1.084152    -0.79   0.429    -2.984617    1.267719
           3 |  -1.226199   1.011079    -1.21   0.225     -3.20906    .7566629
           4 |  -2.433855   1.010772    -2.41   0.016    -4.416114    -.451595
           5 |  -3.277704   1.121079    -2.92   0.003    -5.476291   -1.079116
             |
       _cons |   98.42056   .9392676   104.78   0.000     96.57853    100.2626
------------------------------------------------------------------------------
```

SOURCE "Dohoo" Methods in Epidemiologic Research. Dohoo, I. R., Martin, S. W., & Stryhn, H. (2012). Methods in epidemiologic research. Charlottetown, P.E.I.: VER, Inc. Free online: https://projects.upei.ca/mer/

Example 14.6 Using and interpreting regular indicator variables in linear regression
data = bw5k

A model for -bwt- was fit with -gest-, and 3 indicator variables for mother's education level as the predictors.

bwt	Coef	SE	t	P>t	95% CI	
gest21	124.383	2.938	42.33	0.000	118.622	130.144
meduc_c4=2	20.046	20.980	0.96	0.339	-21.083	61.176
meduc_c4=3	53.270	21.887	2.43	0.015	10.361	96.179
meduc_c4=4	80.599	19.272	4.18	0.000	42.818	118.380
constant	1057.097	53.930	19.60	0.000	951.370	1162.824

In this instance, the referent level is 'less than high school diploma'; the -bwt- of these babies for a 21-week gestation (gest21=0) is 1,057 gm (the intercept). The coefficient for each education level variable reflects the difference in -bwt- between babies born to those mothers and those born to mothers with less than a high school diploma. We can use a multiple partial *F*-test to test the overall significance of the education level predictors (P=0.001). Individually, the 2 highest levels of education are significantly different from the baseline, but having a high school diploma is not.

Table 3

Factors Associated With the Development of Vascular Diseases, Multivariate Analyses Results, Tinea Capitis Study, Israel, 1966–2011

Factor	Ischemic Heart Disease, Stroke, or Carotid Artery Stenosis		Stroke		Carotid Artery Stenosis		Ischemic Heart Disease	
	RR[a]	95% CI	RR[a]	95% CI	RR[a]	95% CI	RR[a]	95% CI
Radiation, yes vs. no	1.19	1.09, 1.29	1.35	1.20, 1.53	1.32	1.06, 1.64	1.12	1.01, 1.26
Sex, female vs. male	0.52	0.47, 0.57	0.60	0.52, 0.68	0.81	0.65, 1.02	0.34	0.29, 0.39
Age, increase of 1 year	1.03	1.02, 1.04	1.04	1.03, 1.06	1.08	1.06, 1.11	1.03	1.02, 1.04
Smoking, yes vs. no	1.54	1.41, 1.68	1.44	1.27, 1.63	2.09	1.66, 2.64	1.85	1.64, 2.09
SES[b], continuous, increment of 1 point	0.96	0.94, 0.99	0.97	0.93, 1.01	0.93	0.86, 1.00	0.94	0.91, 0.98
Hypertension, yes vs. no	2.84	2.58, 3.13	3.72	3.22, 4.29	3.82	2.93, 4.96	3.05	2.69, 3.47
Diabetes, yes vs. no	1.73	1.59, 1.89	2.02	1.79, 2.29	2.09	1.66, 2.62	1.91	11.71, 2.15

Abbreviations: CI, confidence interval; RR, relative risk; SES, socioeconomic status.

[a] Relative risks of each variable are adjusted for all other variables; excess RR = RR − 1.

[b] Based on national residential classification ranging from 1 (low) to 10 (high).

SOURCE:

Siegal Sadetzki, Angela Chetrit, Ben Boursi, Osnat Luxenburg, Ilya Novikov, Arnon Cohen, Childhood Exposure to Low to Moderate Doses of Ionizing Radiation and the Risk of Vascular Diseases: The Tinea Capitis Study, *American Journal of Epidemiology*, Volume 190, Issue 3, March 2021, Pages 423–430, https://doi-org.myaccess.library.utoronto.ca/10.1093/aje/kwaa177

TABLE 2—Simultaneously Adjusted Multivariable Logistic Regression Models of Factors Predicting Young Adult Smoking Cessation Behaviors: Ontario Tobacco Survey, July 2005–December 2008

Characteristic	Model 1: Attempt to Quit (< 30 d) vs No Attempt to Quit, OR (95% CI)[a]	Model 2: Abstinence (≥ 30 d) vs Attempt (< 30 d) and No Attempt to Quit, OR (95% CI)[a]
Age, y	0.97 (0.89, 1.06)	0.94 (0.83, 1.07)
Heaviness of smoking index	0.85 (0.68, 1.06)	0.64* (0.46, 0.89)
Gender		
Women (Ref)	1.00	1.00
Men	1.52 (0.81, 2.87)	0.74 (0.36, 1.52)
Education		
≤ high school (Ref)	1.00	1.00
> high school	0.84 (0.42, 1.67)	1.81 (0.73, 4.50)
Self-efficacy		
Less or uncertain confidence (Ref)	1.00	1.00
Very confident	0.53 (0.26, 1.06)	3.27* (1.15, 9.28)
Use of cessation resources or supports[b]		
No (Ref)	1.00	1.00
Yes	1.86* (1.05, 3.31)	3.46* (1.66, 7.19)
Support to quit[c]		
No (Ref)	1.00	1.00
Yes	0.42 (0.16, 1.10)	4.88* (1.29, 18.44)
Smoking in the home		
People smoke indoors (Ref)	1.00	1.00
No one smokes indoors	1.79 (0.97, 3.32)	1.31 (0.52, 3.27)
Someone make quitting more difficult		
No (Ref)	1.00	1.00
Yes	1.14 (0.57, 2.29)	0.72 (0.34, 1.51)
Prior number of quit attempts		
0 (Ref)	1.00	1.00
1	1.46 (0.51, 4.17)	1.34 (0.45, 4.02)
≥ 2	3.50* (1.28, 9.55)	1.58 (0.54, 4.59)
Quit intention		
No (Ref)	1.00	1.00
Yes	3.94* (2.22, 6.99)	1.48 (0.74, 2.98)
Perceived addiction		
Less addicted (Ref)	1.00	1.00
Very addicted	1.20 (0.57, 2.51)	0.58 (0.26, 1.29)
Smoking medications make quitting a lot easier[d]		
No (Ref)	1.00	1.00
Yes	2.30* (1.16, 4.57)	1.54 (0.60, 3.95)

Note. CI = confidence ratio; OR = odds ratio. Data are weighted.
[a]Variance inflation factor statistics were all satisfactory in both models.
[b]Use of cessation aids and resources include any use of self-help materials, behavioral therapies, pharmacotherapies, physician advice to quit smoking, and participation in a local quit program measured during 6-month follow-up; all other characteristics were measured at the baseline interview.
[c]Support to quit (yes vs no, "don't know") was determined from the question "If you decided to quit smoking, do you have someone you can count on for support?"
[d]Perceived knowledge for the benefits of stop smoking medications and counseling.
*$P < .05$.

Young Adult Smoking Cessation: Predictors of Quit Attempts and Abstinence

Lori M. Diemert, MSc, Susan J. Bondy, PhD, K. Stephen Brown, PhD, and Steve Manske, PhD

About the Authors

Lori M. Diemert is with the Ontario Tobacco Research Unit, University of Toronto, Toronto, Ontario, Canada. Susan

Table 4. Risk of Motor Neurone Disease With Occupational Exposure to Extremely Low-Frequency Magnetic Fields in a Population-Based Case-Control Study, New Zealand, 2013–2016

Exposure to ELF-MF	Cases (*n* = 319)		Controls (*n* = 604)		Age- and Sex-Adjusted Model[a]		Final Model[b]	
	No.	%	No.	%	OR	95% CI	OR	95% CI
Background level	130	41	227	38	1.00	Referent	1.00	Referent
Ever exposed above background level	189	59	377	62	0.87	0.66, 1.15	0.77	0.56, 1.05
Exposure level								
Background level	130	41	227	38	1.00	Referent	1.00	Referent
Medium level only	161	50	326	54	0.87	0.65, 1.16	0.77	0.56, 1.06
Ever exposed at high level	28	9	51	8	0.88	0.52, 1.48	0.71	0.39, 1.28
Duration of exposure, years								
Background level	130	41	227	38	1.00	Referent	1.00	Referent
Exposure <3 years	55	17	99	16	0.95	0.64, 1.42	0.88	0.58, 1.35
Exposure 3–8 years	45	14	93	15	0.82	0.54, 1.25	0.74	0.47, 1.16
Exposure 9–23 years	40	13	92	15	0.78	0.50, 1.21	0.71	0.45, 1.13
Exposure >23 years	49	15	93	16	0.92	0.61, 1.40	0.73	0.46, 1.15
P value (test for trend)					0.85		0.47	
Background level	130	41	227	38	1.00	Referent	1.00	Referent
Medium level[c] <3 years	40	12	77	13	0.91	0.58, 1.43	0.85	0.53, 1.35
Medium level 3–8 years	41	13	79	13	0.89	0.57, 1.38	0.80	0.50, 1.28
Medium level 9–23 years	37	12	86	14	0.79	0.50, 1.23	0.72	0.45, 1.16
Medium level >23 years	43	13	84	14	0.89	0.58, 1.38	0.71	0.45, 1.14
Ever high level[d] <3 years	15	5	22	4	1.07	0.53, 2.17	0.96	0.45, 2.07
Ever high level 3–8 years	4	1	14	2	0.46	0.15, 1.44	0.35	0.10, 1.21
Ever high level 9–23 years	3	1	6	1	0.71	0.17, 2.95	0.50	0.11, 2.27
Ever high level >23 years	6	2	9	1	1.22	0.41, 3.68	0.76	0.24, 2.41
Cumulative exposure, unit-years[e]								
Background level	130	41	227	38	1.00	Referent	1.00	Referent
Exposure <4 unit-years	52	16	96	16	0.93	0.62, 1.41	0.87	0.57, 1.34
Exposure 4–12 unit-years	47	15	106	17	0.77	0.51, 1.16	0.72	0.46, 1.11
Exposure 13–28 unit-years	43	13	83	14	0.93	0.60, 1.43	0.80	0.50, 1.26
Exposure >28 unit-years	47	15	92	15	0.87	0.57, 1.33	0.67	0.41, 1.07
P value (test for trend)					0.96		0.42	

Abbreviations: CI, confidence interval; ELF-MF, extremely low-frequency magnetic fields; OR, odds ratio.

[a] OR adjusted for age and sex.

[b] OR adjusted for age, sex, education, ethnicity, socioeconomic status, smoking status, sports, alcohol, head injury, spine injury, electric shocks, and solvents.

[c] Duration above background (for those with medium exposure only).

[d] Duration above background (for those with ever high exposure).

[e] Cumulative exposure (unit-years) is the product of duration and level of exposure (background level assigned 0, medium level exposure assigned 1, high exposure level assigned 4).

SOURCE:
Chen, G., Mannetje, A., Douwes, J., van den Berg, L., Pearce, N., Kromhout, H., Glass, B., Brewer, N., & McLean, D. (2021). Associations of Occupational Exposures to Electric Shocks and Extremely Low-Frequency Magnetic Fields With Motor Neurone Disease. American Journal of Epidemiology, 190(3), 393–402. https://doi.org/10.1093/aje/kwaa214

Downloaded from https://academic.oup.com/aje/article/190/3/393/5919887 by Bora Laskin Law Library user on 11 March 2021